

Principles of Communication System

Module 3:

* Fundamentals of Frequency Modulation.

- Basic principles of FM.
- Principles of Phase Modulation.
- Modulation index and sidebands.
- Noise Suppression.
- Effects of FM.
- Frequency Modulation
vs
Amplitude Modulation.

* FM Circuits.

- Frequency Modulators.
- Voltage Controlled Oscillators.
- Frequency Demodulators.
- Slope Detectors.
- Phase Locked Loops.

* Communication Receiver

- Super heterodyne receiver.
- Frequency Conversion.
- Mixing Principles.
- JFET Mixer.

Principles of Communication System

Basic Principles of Frequency Modulation:

Frequency modulation: (FM).

* It is a form of angle modulation in which the instantaneous frequency $f_i(t)$ is varied linearly with message signal $m(t)$.

$$f_i(t) = f_c + k_f m(t). \quad \text{--- (1)}$$

where f_c = carrier frequency (unmodulated carrier).

k_f = frequency sensitivity, Hz/volt.

integrating wrt 't' and multiplying 2π for equation (1)

we get.

$$\int_0^t 2\pi f_i(t) dt = \int_0^t 2\pi f_c dt + \int_0^t 2\pi k_f m(t) dt$$

where $\phi_i(t) = \int_0^t 2\pi f_i(t) dt. \Rightarrow \phi_i = \text{Changes by } 2\pi \text{ radian then complete oscillation occurs.}$

$$= \phi_i(t) = 2\pi f_c t + 2\pi k_f \int_0^t m(t) dt.$$

The Frequency Modulated signal is given by

$$s(t) = A_c \cos(\phi_i(t))$$

$$= A_c \cos \left[2\pi f_c t + 2\pi k_f \int_0^t m(t) dt \right] \quad \text{--- (2)}$$

Principles of Communication Systems

Basic Principles of Phase Modulation:

Phase Modulation (PM):

* It is a form of angle modulation in which the instantaneous angle $\phi_i(t)$ is varied linearly with the message signal.

$$\phi_i(t) = 2\pi f_c t + k_p m(t).$$

where $2\pi f_c t \rightarrow$ angle of unmodulated carrier.

$k_p \rightarrow$ phase sensitivity of modulator, rad/volt.

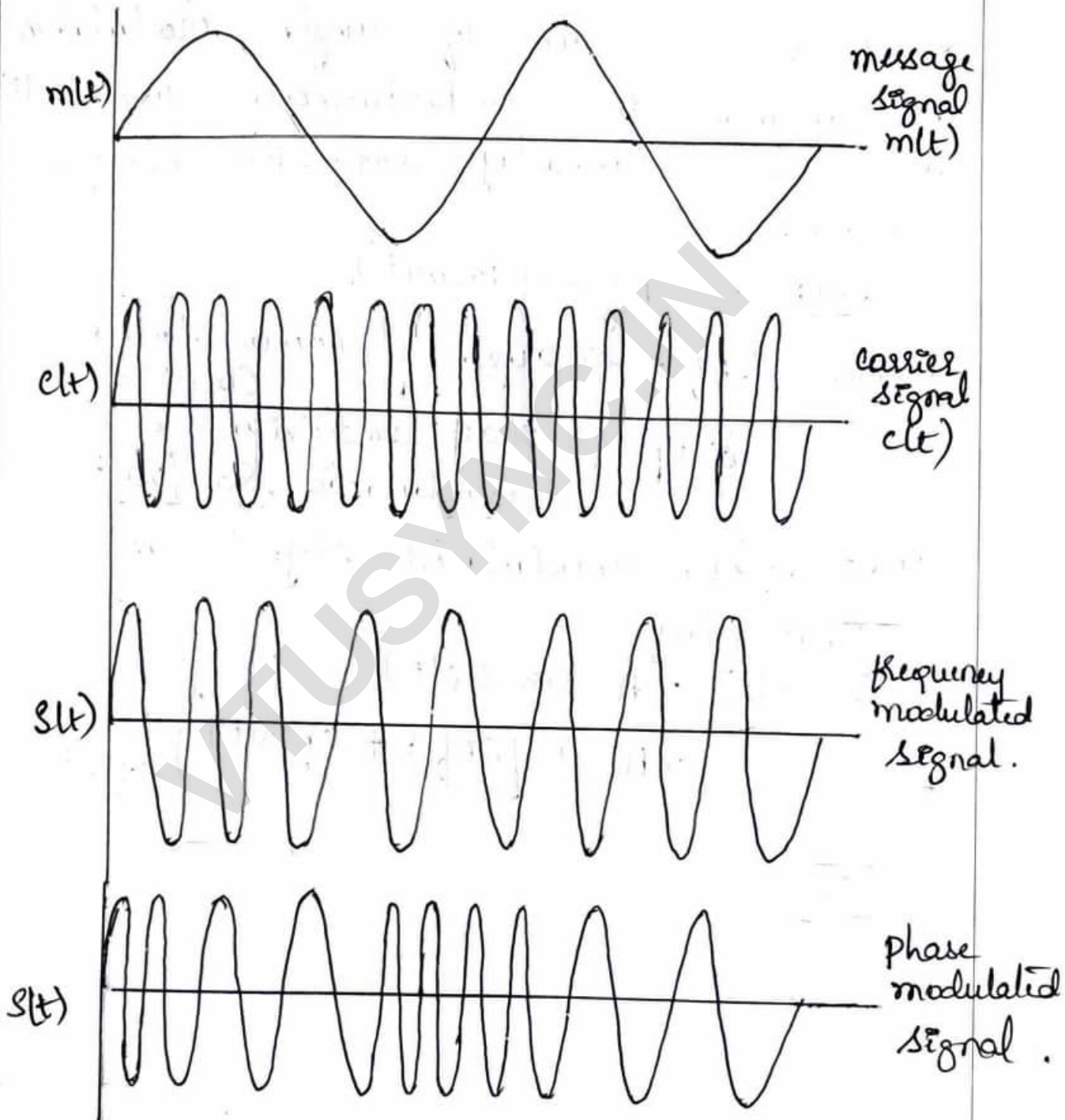
∴ phase modulated signal is given by.

$$s(t) = A_c \cos \phi_i(t).$$

$$= A_c \cos [2\pi f_c t + k_p m(t)] \quad \text{--- (1)}$$

Principles of Communication Systems

* wave form for frequency & phase modulated signal is, as shown below:



Principles of Communication Systems

Spectrum analysis of FM signal.

* Let us consider sinusoidal modulating signal

$$m(t) = A_m \cos 2\pi f_m t \quad \text{--- (1)}$$

* The instantaneous frequency

$$f_i(t) = f_c + k_f m(t) \quad \text{--- (2)}$$

(1) in (2)

$$f_i(t) = f_c + k_f A_m \cos 2\pi f_m t$$

$$f_i(t) = f_c + f_d \cos 2\pi f_m t \quad (\Delta f = f_d = \text{frequency deviation})$$

$$\text{where } \Delta f = f_d = k_f A_m$$

The quantity f_d is frequency deviation, which is proportional to A_m .

Modulation index: m_f or β .

* The ratio of the frequency deviation to the modulating frequency is known as the modulation index m_f or β .

$$\beta = m_f = \frac{f_d}{f_m} = \frac{\text{frequency deviation}}{\text{modulating frequency}}$$

Principles of Communication Systems

Problems:

① what is the deviation ratio of TV sound if the maximum deviation is 25 kHz and the maximum modulating frequency is 15 kHz?

Solution: $f_d = 25 \text{ kHz}$, $f_m = 15 \text{ kHz}$; $m_f = ?$

$$m_f = \frac{f_d}{f_m} = \frac{25 \text{ K}}{15 \text{ K}} = \underline{\underline{1.667}}$$

② The transmitter operates on a frequency of 915 MHz. The maximum FM deviation is $\pm 12.5 \text{ kHz}$. what are maximum and minimum frequencies that occur during modulation?

Solution: $915 \text{ MHz} = 915,000 \text{ kHz}$.

$$\begin{aligned} \text{Maximum deviation} &= 915,000 + 12.5 \\ &= \underline{\underline{915,012.5 \text{ kHz}}} \end{aligned}$$

$$\begin{aligned} \text{Minimum deviation} &= 915,000 - 12.5 \\ &= \underline{\underline{914,987.5 \text{ kHz}}} \end{aligned}$$

Principles of Communication System

Depending on value of modulation index β , we classify FM as.

- 1) Narrow-band FM: β is small compared to one radian
- 2) wide-band FM: β is large compared to one radian

wide band FM wave:

- The FM wave for sinusoidal modulation is given by

$$s(t) = A_c \cos [2\pi f_c t + \beta \sin 2\pi f_m t] \quad \text{--- (1)}$$

$$\text{Let } \theta = 2\pi f_c t + \beta \sin 2\pi f_m t.$$

Equation (1) is having only real part.

$$s(t) = \operatorname{Re} [A_c e^{j\theta}]$$

$$= \operatorname{Re} [A_c e^{j(2\pi f_c t + \beta \sin 2\pi f_m t)}]$$

$$s(t) = \operatorname{Re} [A_c e^{j2\pi f_c t} \cdot e^{j\beta \sin 2\pi f_m t}]$$

$$= \operatorname{Re} [e^{j2\pi f_c t} \cdot A_c e^{j\beta \sin 2\pi f_m t}]$$

$$\text{where } A_c e^{j\beta \sin 2\pi f_m t} = \hat{s}(t) \quad \text{--- (2)}$$

$$s(t) = \operatorname{Re} [e^{j2\pi f_c t} \cdot \hat{s}(t)] \quad \text{--- (3)}$$

where $\hat{s}(t)$ is a periodic time function with a fundamental frequency f_m . This can be expressed using complex Fourier series as,

$$\hat{s}(t) = \sum_{n=-\infty}^{\infty} c_n e^{j2\pi n f_m t} \quad \text{--- (4)}$$

Principles of Communication System

where C_n is a complex fourier co-efficient given by.

$$C_n = \frac{1}{T_m} \int_{-\frac{1}{2}T_m}^{\frac{1}{2}T_m} \hat{s}(t) e^{-j2\pi n f_m t} \cdot dt \quad \text{--- (5)}$$

Substituting (2) in (5).

$$C_n = \frac{1}{T_m} \int_{-\frac{1}{2}T_m}^{\frac{1}{2}T_m} A c e^{j\beta \sin 2\pi f_m t} \cdot e^{-j2\pi n f_m t} dt.$$

$$C_n = A c b_m \int_{-\frac{1}{2}T_m}^{\frac{1}{2}T_m} e^{j[\beta \sin(2\pi f_m t) - \underbrace{2\pi n f_m t}_{\text{---}}]} dt.$$

Let $x = 2\pi f_m t$
differentiating wrt. t .

$$\frac{dx}{dt} = 2\pi f_m.$$

$$\boxed{dt = \frac{dx}{2\pi f_m}}$$

$$x = 2\pi f_m t$$

when $t = -\frac{1}{2f_m}$; when $t = \frac{1}{2f_m}$

$$x = 2\pi f_m \left(-\frac{1}{2f_m}\right) ; x = 2\pi f_m \left(\frac{1}{2f_m}\right)$$

$$\boxed{x = -\pi} ; \boxed{x = \pi}$$

$$C_n = A c b_m \int_{-\pi}^{\pi} e^{j(\beta \sin x - nx)} \cdot \frac{dx}{2\pi f_m}$$

$$C_n = \frac{A c}{2\pi} \int_{-\pi}^{\pi} e^{j(\beta \sin x - nx)} dx.$$

Principles of Communication System

$$C_n = A_c J_n(\beta) \quad \text{--- (6)}$$

where $J_n(\beta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j(\beta \sin x - nx)} dx$.

$J_n(\beta)$ is a Bessel function of the 1st kind, n th order with an argument β .

Sub (6) in (4).

$$\hat{s}(t) = \sum_{n=-\infty}^{\infty} A_c J_n(\beta) e^{j2\pi n f_m t} \quad \text{--- (7)}$$

using (7) in (3)

$$s(t) = \operatorname{Re} \left[e^{j2\pi f_c t} \cdot \sum_{n=-\infty}^{\infty} A_c J_n(\beta) e^{j2\pi n f_m t} \right]$$

$$= \operatorname{Re} \left[A_c \sum_{n=-\infty}^{\infty} J_n(\beta) e^{j2\pi (f_c + n f_m) t} \right]$$

where $e^{j\theta} = \cos\theta + j\sin\theta$. (Trigonometric identities).

Real part. $\operatorname{Re}[e^{j\theta}] = \cos\theta$. $\theta = 2\pi (f_c + n f_m) t$.

$$s(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta) \cos(2\pi (f_c + n f_m) t) \quad \text{--- (8)}$$

Given the values for n blw $-\infty$ to ∞ .
i.e. $n = 0, 1, -1, 2, -2, \dots, +\infty, -\infty$.

Principles of Communication System

$$S(t) = A_c \left[J_0(\beta) \cos 2\pi f_c t + J_1(\beta) \cos (2\pi (f_c + f_m)t) + J_{-1}(\beta) \cos 2\pi (f_c - f_m)t + J_2(\beta) \cos 2\pi (f_c + 2f_m)t + J_{-2}(\beta) \cos 2\pi (f_c - 2f_m)t + \dots \right] \quad \text{--- (9)}$$

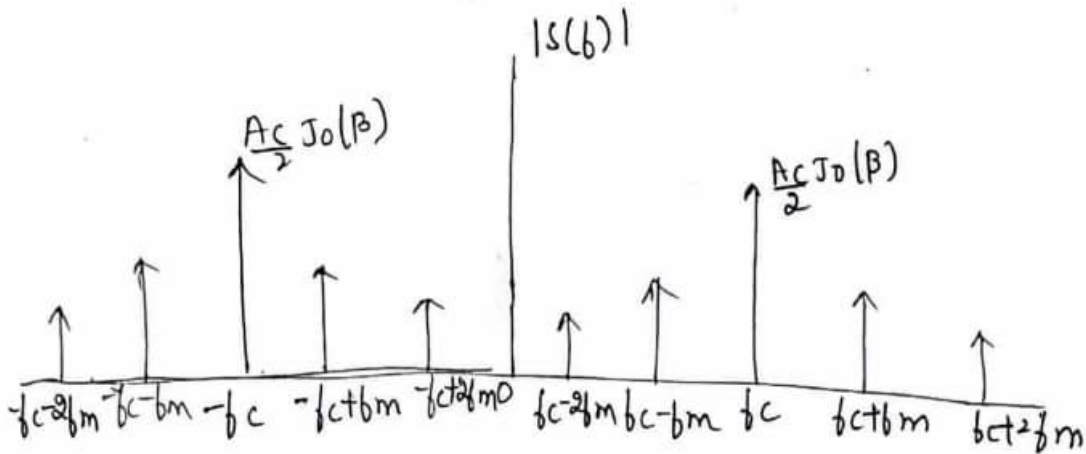
$$S(t) = A_c \left[J_0(\beta) \cos 2\pi f_c t + J_1(\beta) [\cos (2\pi (f_c + f_m)t) - \cos 2\pi (f_c - f_m)t] + J_2(\beta) [\cos 2\pi (f_c + 2f_m)t - \cos 2\pi (f_c - 2f_m)t] + \dots \right]$$

Thus the modulated signal has a carrier component and an infinite number of side frequencies $f_c \pm f_m, f_c \pm 2f_m, \dots, f_c \pm n f_m$.

Taking fourier transform on both sides of eq (9).

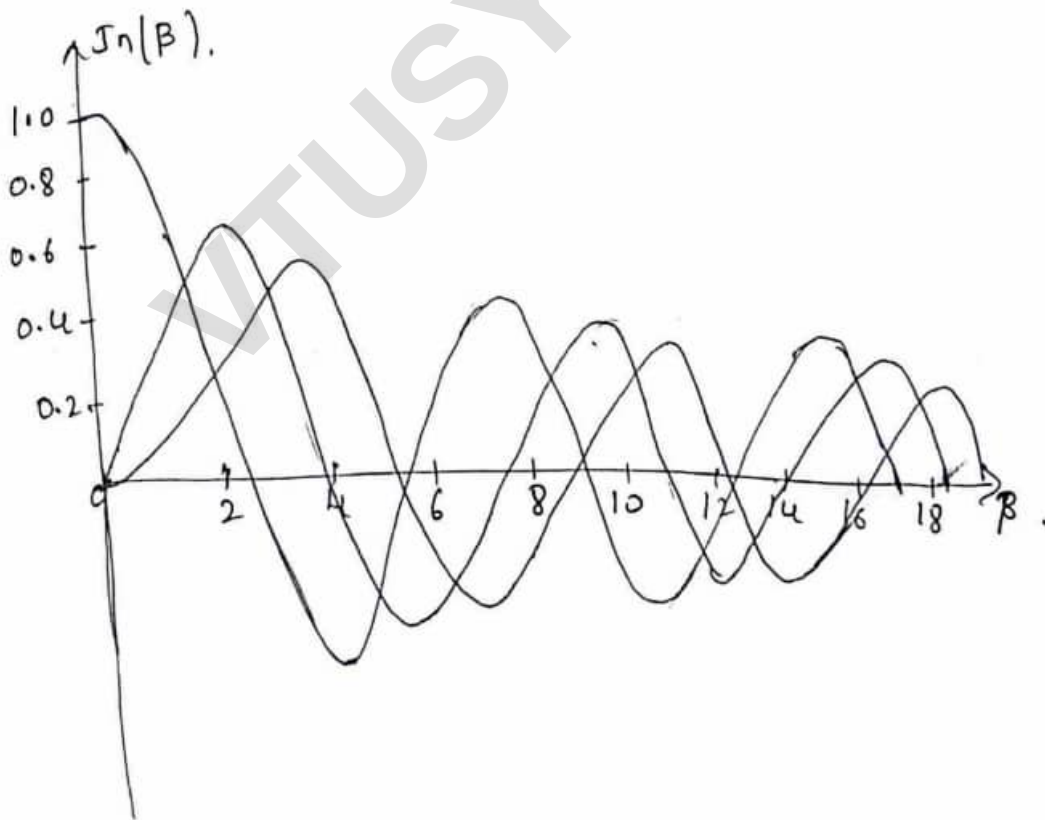
$$S(f) = \frac{A_c}{2} J_0(\beta) [\delta(f - f_c) + \delta(f + f_c)] + \frac{A_c}{2} J_1(\beta) [\delta(f - (f_c + f_m)) + \delta(f + (f_c + f_m))] + \frac{A_c}{2} J_{-1}(\beta) [\delta(f - (f_c - f_m)) + \delta(f + (f_c - f_m))] + \dots$$

Principles of Communication System



The amplitude of side frequency component depends upon the Bessel function.

The Bessel variations as a function of β fixing the values of 'n' as shown in figure below.



Principles of Communication System

Transmission Bandwidth of FM signals.

- * In theory, an FM signal contains an infinite number of side frequencies. So bandwidth required to transmit is infinite in extent.
- * In practical the FM signal is effectively limited to a finite number of significant side frequencies.
- * Actually for a side frequencies which are separated from center frequency f_c by an amount frequency deviation Δf , will decay rapidly to zero.
- * For large values of modulation index ' β ' the total bandwidth is slightly greater than $2\Delta f$.
- * For smaller values of ' β ', the spectrum is limited to f_c and one pair of side frequency $f_c \pm f_m$. So bandwidth approach is $2f_m$.

$$\therefore B_T \approx 2\Delta f + 2f_m.$$

$$= 2\Delta f \left(1 + \frac{f_m}{\Delta f}\right) \quad \because \beta = \frac{\Delta f}{f_m}.$$

$$B_T = 2\Delta f \left(1 + \frac{1}{\beta}\right) \dots \text{Carson's rule.}$$

Principles of Communication System

We retain the max number of significant side frequencies whose amplitude are all greater than 1% of unmodulated carrier amplitude.

Transmission bandwidth of a FM wave as the separation b/w the two frequencies beyond which none of the side frequencies is greater than 1% of the carrier amplitude obtained when the modulation is removed.

We define transmission bandwidth as:

$$BW = 2f_m N$$

where - N is the number of significant sidebands in the signal.

- f_m is frequency of message signal.

* Another way to determine the bandwidth of an FM signal is to use Carson's rule.

$$\text{ie } BW = 2[f_{d(\max)} + f_m(\max)]$$

Principles of Communication System

Problems:

① what is the maximum bandwidth of an FM signal with a deviation of 30 kHz and a maximum modulating signal of 5 kHz. determine.

① modulation index.

② BW by carson's rule.

Solution:

Given $\Delta f = f_d = 30 \text{ kHz}$.

$f_m = 5 \text{ kHz}$.

a) modulation index:

$$m_f = \frac{f_d}{f_m} = \frac{30 \text{ k}}{5 \text{ k}} = 6.$$

② carson's rule:

$$BW = 2 [f_{d\max} + f_{m\max}]$$

$$= 2 [30 \text{ k} + 5 \text{ k}]$$

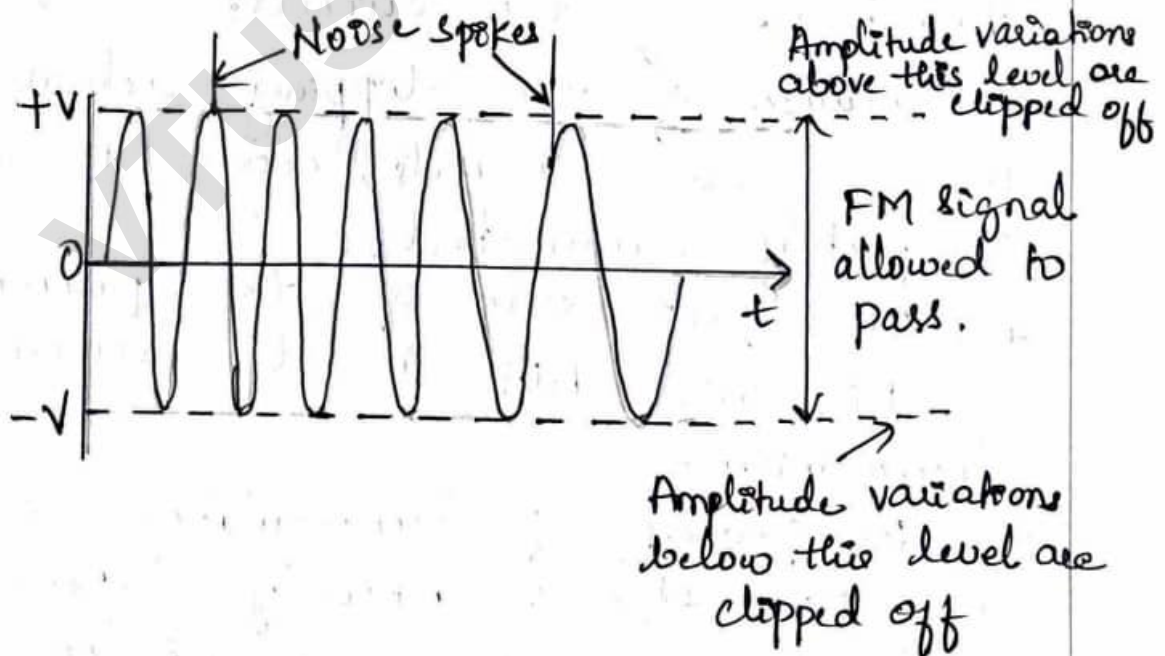
$$= 2 [35 \text{ k}]$$

$$\boxed{BW = 70 \text{ kHz}}$$

Principles of Communication System

Noise Suppression effects of FM:

- * Noise is interference generated by lightning, motors, automotive ignition systems, and any power line switching that produce transient signals.
- * Such noise is typically narrow spikes of voltage with very high frequencies.
- * They add to a signal and interfere with it.
- * The potential effect of such noise on an FM signal is shown in fig:



Principles of Communication System

- * FM signals, however have a constant modulated carrier amplitude, and FM receivers contain limiter circuits that deliberately restrict the amplitude of the received signal.
- * Any amplitude variations occurring on the FM signal are effectively clipped off, as shown in fig.
- * This does not effect the information content of the FM signal, since it is contained solely within the frequency variations of the carrier.
- * Because of the clipping actions of the limiter circuits, noise is almost completely eliminated.
- * In fact, one of the primary benefits of FM over AM is its superior noise immunity.
- * The process of demodulating & recovering an FM signal actually suppresses noise and improves the signal-to-noise ratio.

Principles of Communication System

Frequency Modulation Versus Amplitude modulation.

Amplitude Modulation	Frequency modulation
1) Amplitude of AM is not constant.	1) Amplitude of FM signal is constant.
2) Value of modulation index is one or zero	2) Value of modulation index is less or more than one.
3) The signal bandwidth is twice the high modulating frequency	3) The signal frequency is equal to the modulation index.
4) Transmitter cost is High.	4) Transmitter cost is medium.
5) Amplitude of carrier signal changes	5) Frequency of carrier signal changes
6) AM is used for broadcasting in long wave or medium wave or short wave bands.	6) FM can be used for the broadcasting of FM radio. This helps in larger signal to noise ratio

Principles of Communication System

Advantages and Disadvantages of Amplitude Modulation, and applications of AM.

Advantages of AM:

- ① AM transmitter are less complex.
- ② AM receivers are simple and detection is easy.
- ③ AM waves can travel longer distance.
- ④ Low bandwidth.

Disadvantage of AM:

- ① Requires large power.
- ② Gets affected due to noise.
- ③ Requires large bandwidth.

Applications of AM:

- ① Radio broadcasting
- ② picture transmission in TV.

Principles of Communication System

Advantages, disadvantages and applications of FM.

Advantages of FM:

- 1) Transmitted power remains constant.
- 2) FM receivers are immune to noise.
- 3) Good capture effect.
- 4) No mixing of signals.

Disadvantages of FM:

- 1) It uses too much spectrum space.
- 2) The bandwidth is wider.
- 3) The modulation index can be kept low to minimize the bandwidth used.

Applications of FM:

- ① FM radio & broadcasting.
- ② Sound transmission in TV.
- ③ police wireless.

Principles of Communication System

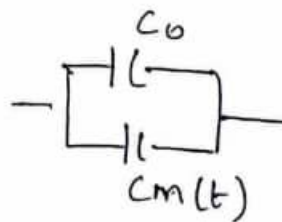
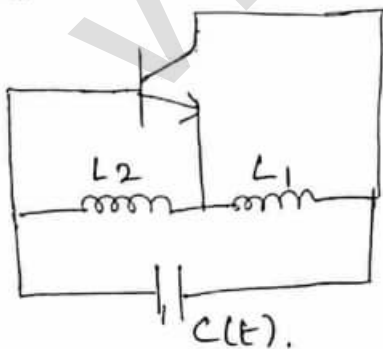
Frequency Modulators:

- * A frequency modulator is a circuit that varies carrier frequency in accordance with the modulating signal.
- * The carrier is generated by either an LC or a crystal oscillator circuit.



Generation of FM signals:

- In direct FM system, the instantaneous frequency of the carrier wave is varied directly in accordance with the message signal by means of a device called voltage controlled oscillator (VCO).
- Hartley oscillator is used which produces high stable sinusoidal oscillations as shown. The capacitive component of the frequency determining n/w in the oscillator consists of a fixed capacitor paralleled by voltage-variable capacitor, also called a varactor or varicap.
- The resultant capacitance is represented by $C(t)$ as shown in fig.



PRINCIPLES OF COMMUNICATION SYSTEMS

- The varactor is a variable voltage capacitor, obtained by a p-n junction diode that is reversed bias.
- The frequency of oscillation of Hartley oscillator is

$$f_i(t) = \frac{1}{2\pi\sqrt{(L_1+L_2)C(t)}} \quad \text{--- (1)}$$

where $C(t) = C_0 + \Delta C \cos(2\pi f_m t)$.

where C_0 is total capacitance in absence of modulation.
 ΔC is max change in capacitance.

$$f_i(t) = \frac{1}{2\pi\sqrt{(L_1+L_2)(C_0 + \Delta C \cos(2\pi f_m t))}}$$

$$= \frac{1}{2\pi\sqrt{(L_1+L_2)C_0\left(1 + \frac{\Delta C}{C_0} \cos(2\pi f_m t)\right)}}$$

$$f_i(t) = \frac{1}{2\pi\sqrt{(L_1+L_2)C_0} \sqrt{1 + \frac{\Delta C}{C_0} \cos(2\pi f_m t)}}$$

$$\text{where } f_0 = \frac{1}{2\pi\sqrt{(L_1+L_2)C_0}}$$

$$f_i(t) = f_0 \left[1 + \frac{\Delta C}{C_0} \cos(2\pi f_m t) \right]^{-1/2} \quad \text{--- (2)}$$

PRINCIPLES OF COMMUNICATION SYSTEMS

using Binomial expansion $[1+x]^{-1/2} = 1 - \frac{x}{2!}$

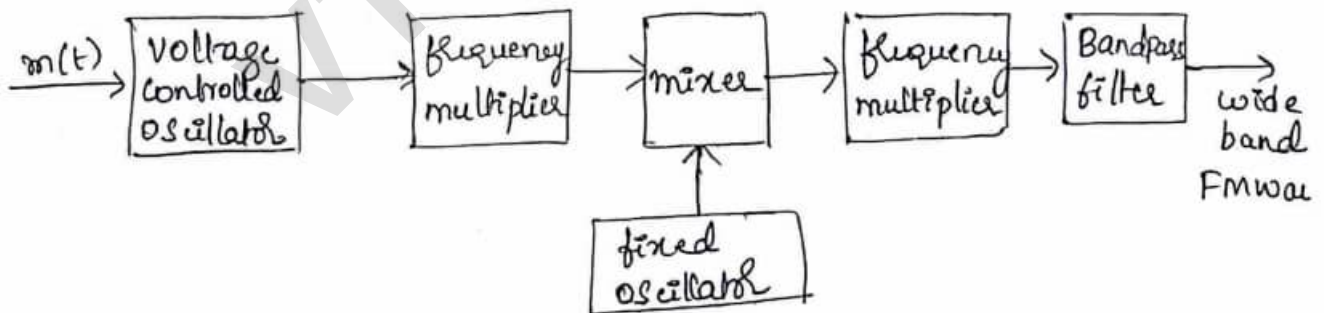
$$f_i(t) \approx f_0 \left[1 - \frac{\Delta C}{2C_0} \cos(2\pi f_m t) \right]$$

$$= f_0 - \frac{\Delta C f_0}{2C_0} \cos(2\pi f_m t).$$

Let $\Delta f = -\frac{\Delta C f_0}{2C_0}$.

$$f_i(t) = f_0 + \Delta f \cos(2\pi f_m t)$$

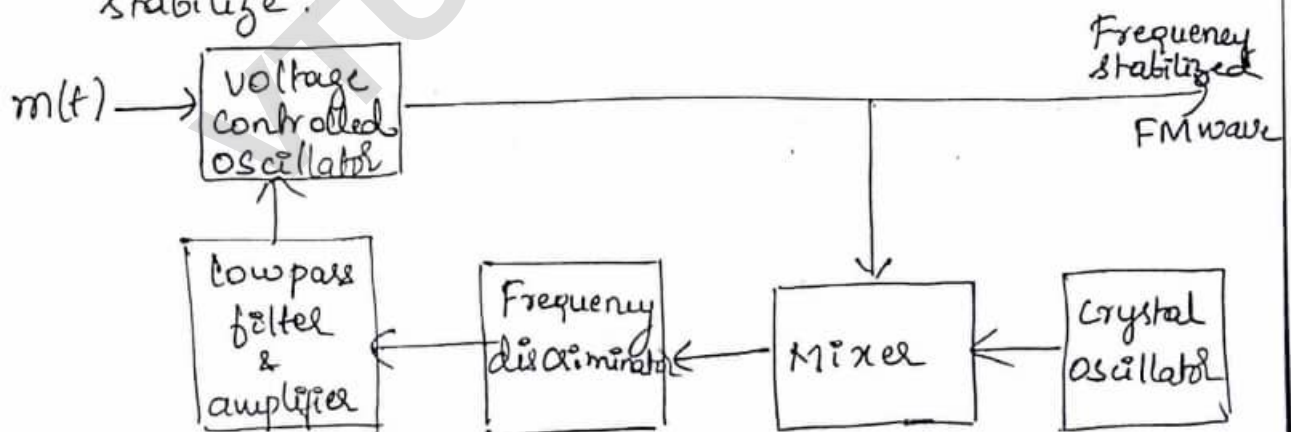
In order to generate a WBFM wave with the required frequency deviation, a configuration as shown in figure is used which consist of VCO followed by a series of frequency multipliers and mixers.



Block diagram of WBFM using a Voltage-controlled oscillator. (direct method)

PRINCIPLES OF COMMUNICATION SYSTEMS

- An FM transmitter using the direct method as described have disadvantages that the carrier frequency is not obtained from a highly stable oscillator. (i.e. carrier frequency is not stable. So to overcome this a negative feedback system is used, in which o/p of FM generator is applied to a mixer together with the o/p of crystal controlled oscillator & ~~frequency~~ the difference frequency term is extracted.
- The mixer o/p is next applied to frequency discriminator and then through low pass filter and amplifier to get a DC signal for VCO to modify the error in frequency & stabilize.

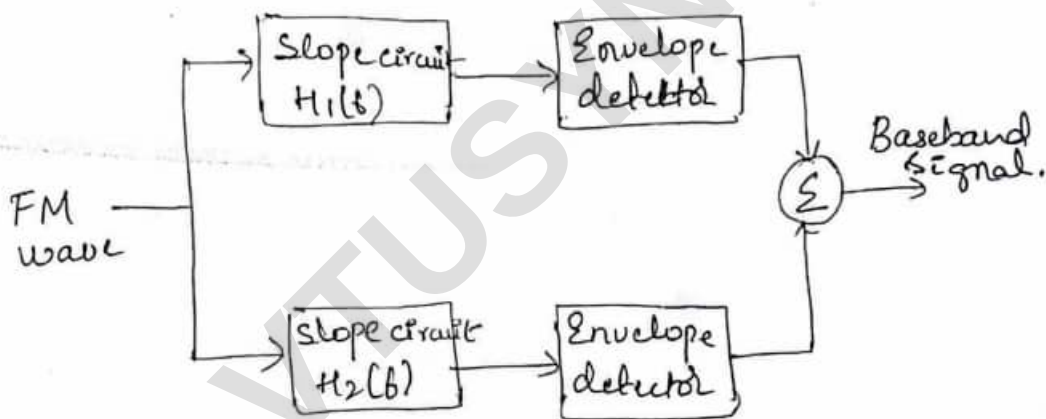


A feedback scheme for frequency stabilization of FM.

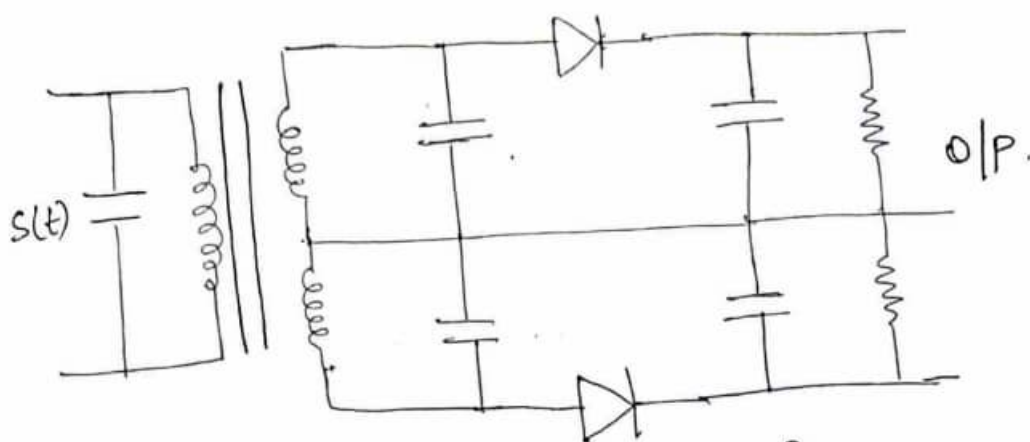
PRINCIPLES OF COMMUNICATION SYSTEMS

Demodulation of FM signal:

- Frequency demodulation is the process of recovering original modulating signal from a modulated signal.
- The objective is to produce a transfer characteristic that is inverse of frequency modulator.
- A direct method of frequency discriminator is used which contains slope circuit and envelope detector.



Balanced frequency discriminator Block diagram.



Balanced slope detector.

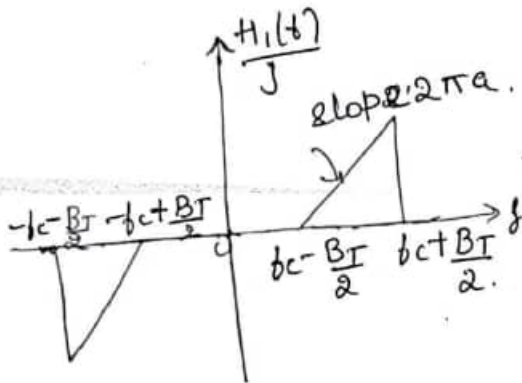
The ideal slope circuit transfer function is given by.

$$H_1(f) = \begin{cases} j2\pi a \left(f - f_c + \frac{B_T}{2}\right) & ; f_c - \frac{B_T}{2} \leq f \leq f_c + \frac{B_T}{2} \\ j2\pi a \left(f + f_c - \frac{B_T}{2}\right) & ; -f_c - \frac{B_T}{2} \leq f \leq -f_c + \frac{B_T}{2} \\ 0 & \text{elsewhere.} \end{cases} \quad \text{--- (1)}$$

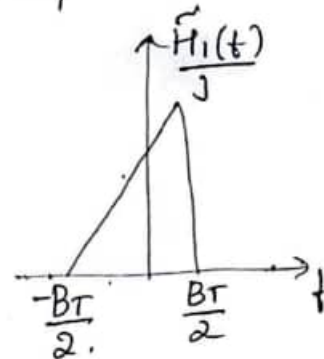
Let $S_1(t)$ be o/p of slope circuit, which is in an interval $f_c - \frac{B_T}{2} \leq |f| \leq f_c + \frac{B_T}{2}$.

PRINCIPLES OF COMMUNICATION SYSTEMS

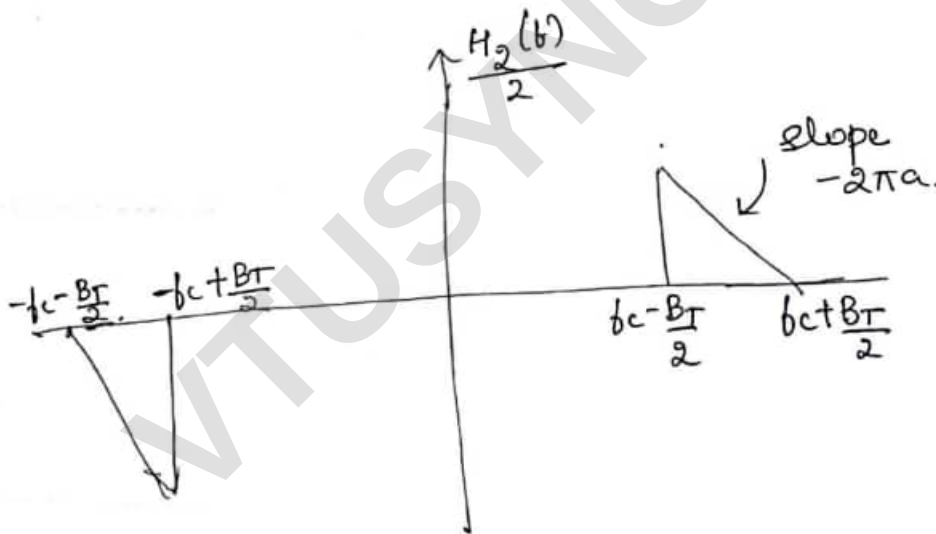
To evaluate the response we consider equivalent lowpass filter with complex envelope.



frequency response
of ideal slope circuit.



slope circuit
response.



complementary complex low pass filter.

PRINCIPLES OF COMMUNICATION SYSTEMS

Let $\tilde{H}_1(f)$ be complex transfer function of slope circuit, this complex transfer function is related to $H_1(f)$ by.

$$\tilde{H}_1(f-f_c) = 2H_1(f), \quad f > 0. \quad \text{--- (2)}$$

using (1) in (2).

$f_c > 0$.

$$\tilde{H}_1(f) = \begin{cases} 2 \int j2\pi a(f + \frac{B_T}{2}) & -\frac{B_T}{2} \leq f \leq +\frac{B_T}{2} \\ 0 & \text{elsewhere} \end{cases} \quad \text{--- (3)}$$

The incoming FM signal,

$$s(t) = A_c \cos \left[2\pi f_c t + 2\pi k_f \int_0^t m(t) dt \right] \quad \text{--- (4)}$$

$$\text{Complex envelope } \tilde{s}(t) = A_c \exp \left[j2\pi k_f \int_0^t m(t) dt \right] \quad \text{--- (5)}$$

Fourier transform of $\tilde{s}_1(t)$ is

$$\tilde{S}_1(f) = \frac{1}{2} \tilde{H}_1(f) \cdot \tilde{S}(f).$$

using eq (3)

$$\tilde{S}_1(f) = \frac{1}{2} \begin{cases} j4\pi a(f + \frac{B_T}{2}) \tilde{S}(f) & -\frac{B_T}{2} \leq f \leq +\frac{B_T}{2} \\ 0 & \text{elsewhere} \end{cases}$$

$$= \begin{cases} j2\pi a(f + \frac{B_T}{2}) \tilde{S}(f) & -\frac{B_T}{2} \leq f \leq +\frac{B_T}{2} \\ 0 & \text{elsewhere} \end{cases}$$

PRINCIPLES OF COMMUNICATION SYSTEMS

But multiplication of Fourier transform of a signal by factor $j2\pi f$ is equivalent to differentiating.

$$\therefore \tilde{S}_1(t) = a \left[\frac{d\tilde{S}(t)}{dt} + j\pi B_T \tilde{S}(t) \right]$$

$$= a \frac{d\tilde{S}(t)}{dt} + j\pi B_T a \tilde{S}(t).$$

Substituting eq (5) we get.

$$\tilde{S}_1(t) = a \left[A_c \frac{d}{dt} \exp \left(j2\pi k_f \int_0^t m(t) dt \right) + j\pi B_T a A_c \exp \left(j2\pi k_f \int_0^t m(t) dt \right) \right]$$

$$= A_c a j2\pi k_f m(t) \exp \left(j2\pi k_f \int_0^t m(t) dt \right) + j\pi B_T a A_c \exp \left(j2\pi k_f \int_0^t m(t) dt \right).$$

common factor.

$$\tilde{S}_1(t) = A_c a j2\pi k_f \left[1 + \frac{2k_f}{B_T} m(t) \right] \exp \left(j2\pi k_f \int_0^t m(t) dt \right) \quad (5)$$

The desired response of slope circuit is with additional phase.

$$S_1(t) = \text{Re} \left[\tilde{S}_1(t) \exp(j2\pi f_c t) \right]$$

PRINCIPLES OF COMMUNICATION SYSTEMS

The signal $s_1(t)$ is a hybrid-modulated signal, in which both amplitude & frequency of the carrier wave vary with message signal $m(t)$.
So we choose.

$$\left| \frac{2K_f}{BT} m(t) \right| < 1 \text{ for all } t$$

Thus the resulting o/p of envelope detector is $|\tilde{s}_1(t)| = \pi B T a A_c \left[1 + \frac{2K_f}{BT} m(t) \right]$.

114 the o/p of second envelope detector is complementary i.e. $\tilde{H}_2(t) = \tilde{H}_1(-t)$.

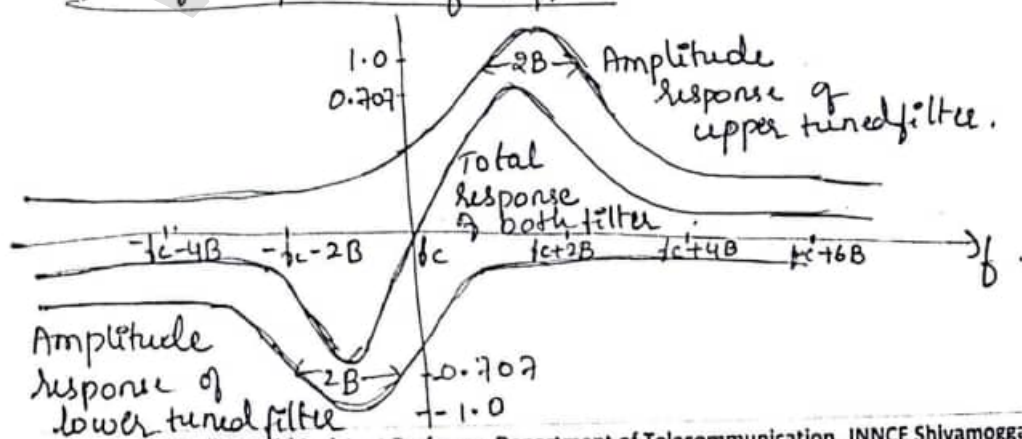
$$\therefore |\tilde{s}_2(t)| = \pi B T a A_c \left[1 - \frac{2K_f}{BT} m(t) \right]$$

The over all o/p.

$$s_o(t) = |\tilde{s}_1(t) - \tilde{s}_2(t)|$$

$$= 4\pi K_f a A_c m(t).$$

Frequency response of o/p:

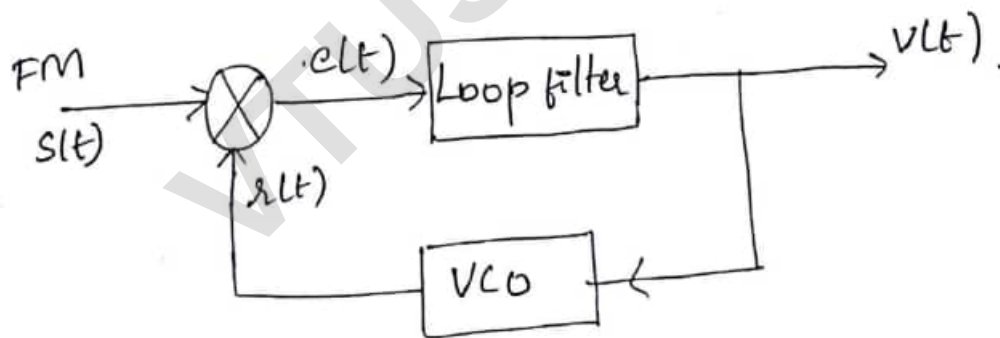


Rashmi M Hullamani | Assistant Professor, Department of Telecommunication, JNNCE Shivamogga

30

Phase Locked Loop (PLL).

- PLL is a negative feedback system. It can be used to track the phase and the frequency of the carrier component of an incoming signal.
- Hence it is useful device for the demodulation of angle modulated signals, when signal to noise ratio is poor.
- A PLL has 3 basic components.
 - 1) A voltage controlled oscillator [VCO]
 - 2) A multiplier serving as a phase detector
 - 3) A loop filter, which is a low pass filter.



- The frequency of VCO can be controlled by the external voltage.
- In a VCO, the oscillation frequency varies linearly with its i/p voltage.

PRINCIPLES OF COMMUNICATION SYSTEMS

- The o/p of the multiplier is passed through a low pass filter and then applied to the i/p of the VCO. This voltage then changes the frequency of the VCO & keeps the loop in locked condition.

~~Initially the control voltage etc~~

- Initially we have adjusted the VCO so that when the control voltage is zero, two conditions are satisfied.

1) The frequency of the VCO is exactly made equal to the unmodulated carrier frequency f_c .

2) The VCO o/p has a 90° phase shift wrt to the unmodulated carrier wave.

- For an i/p signal applied to the phase-locked loop is a FM signal defined by

$$s(t) = A_c \sin [2\pi f_c t + \phi(t)] \quad \text{--- (1)}$$

where A_c is the carrier amplitude, with a modulating signal $m(t)$.

* the angle $\phi_1(t)$ is related to $m(t)$ by the integral

$$\phi_1(t) = 2\pi k_f \int m(t) dt \quad \text{--- (2)}$$

where k_f = frequency sensitivity of

Rashmi M Hullamani | Assistant Professor, Department of Telecommunication, JNNCE Shivamogga

the frequency modulator.

(32)

PRINCIPLES OF COMMUNICATION SYSTEMS

- Let the VCO o/p in the phase-locked loop be defined by.

$$r(t) = A_v \cos [2\pi f_c t + \phi_2(t)] \quad \text{--- (3)}$$

where A_v = amplitude.

If $v(t)$ is control voltage applied to VCO, then angle $\phi_2(t)$ is related to $v(t)$ by the integral

$$\phi_2(t) = 2\pi K_V \int_0^t v(t) dt. \quad K_V = \text{frequency sensitivity of the VCO.}$$

If $s(t)$ and $r(t)$ are 2 i/p's to multiplier, the o/p $e(t)$ will be.

$$e(t) = K_m [s(t)] [r(t)] \quad \text{--- (4)}$$

where K_m = multiplier gain.

Substituting (1) & (3) in (4).

$$e(t) = K_m [A_c \sin [2\pi f_c t + \phi_1(t)]] [A_v \cos [2\pi f_c t + \phi_2(t)]]$$

$$e(t) = K_m A_c A_v \left[\sin (2\pi f_c t + \phi_1(t)) \cos [2\pi f_c t + \phi_2(t)] \right]$$
$$\sin A \cdot \cos B = \frac{1}{2} [\sin (A+B) + \sin (A-B)]$$

$$e(t) = \frac{K_m A_c A_v}{2} \left[\sin [2\pi f_c t + \phi_1(t) + 2\pi f_c t + \phi_2(t)] + \sin [2\pi f_c t + \phi_1(t) - 2\pi f_c t - \phi_2(t)] \right]$$

$$e(t) = \frac{K_m A_c A_v}{2} \left[\sin (4\pi f_c t + \phi_1(t) + \phi_2(t)) + \sin (\phi_1(t) - \phi_2(t)) \right]$$

This is i/p to loop filter. which will pass only 2nd term which is low frequency component

Rashmi M Hullamani | Assistant Professor, Department of Telecommunication, JNNCE Shivamogga

(33)

PRINCIPLES OF COMMUNICATION SYSTEMS

~~addition~~
Non linear model of the phase-locked loop.

wkt.

$$e(t) = \frac{1}{2} K_m A_c A_v \left[\sin(4\pi f_c t + \phi_1(t) + \phi_2(t)) + \sin(\phi_1(t) - \phi_2(t)) \right]$$

It is two components.

- 1) A high frequency component represented by the double-frequency term.

$$K_m A_c A_v \sin(4\pi f_c t + \phi_1(t) + \phi_2(t))$$

- 2) A low-frequency component represented by the difference-frequency term.

$$K_m A_c A_v \sin[\phi_1(t) - \phi_2(t)]$$

The loop filter in the phase-lock loop is a low pass filter, and its response to the high frequency component will be negligible. The VCO also contributes to the attenuation of this component. Therefore, discarding the high-frequency component.

- The i/p to the loop filter is reduced to

$$e(t) = K_m A_c A_v \sin[\phi_1(t) - \phi_2(t)] \\ = K_m A_c A_v \sin[\phi_e(t)]$$

Rashmi M Hullamani | Assistant Professor, Department of Telecommunication, JNNCE Shivamogga

(34)

PRINCIPLES OF COMMUNICATION SYSTEMS

where $\phi_e(t)$ is phase error.

$$\phi_e(t) = \phi_1(t) - \phi_2(t) \quad ; \text{substituting for } \phi_2(t)$$

$$\phi_e(t) = \phi_1(t) - 2\pi k_v \int_0^t v(\tau) d\tau$$

The loop filter operates on i/p $e(t)$ to produce the o/p. defined by convolution integral.

$$v(t) = \int_{-\infty}^{\infty} e(\tau) h(t-\tau) d\tau$$

$$\therefore \phi_e(t) = \phi_1(t) - 2\pi k_v \int_0^t \int_{-\infty}^{\infty} e(\tau) h(t-\tau) d\tau d\tau$$

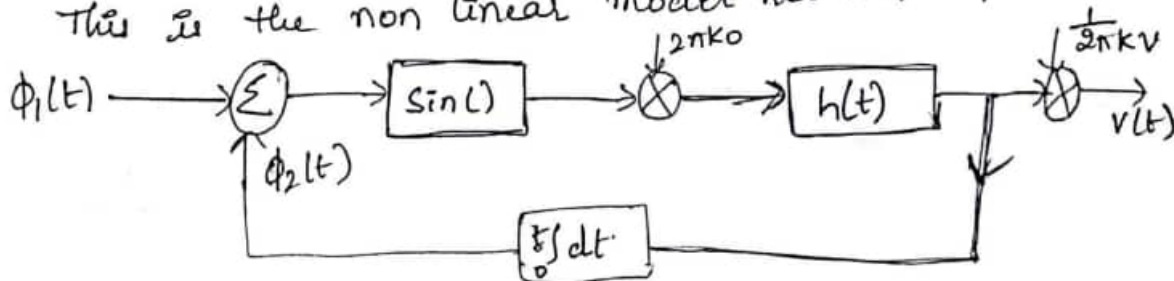
Differentiating w.r.t. t .

$$\frac{d\phi_e(t)}{dt} = \frac{d\phi_1(t)}{dt} - 2\pi k_v k_m A_c A_v \int_{-\infty}^{\infty} \sin(\phi_e(t)) h(t-\tau) d\tau$$

$$\therefore \frac{d\phi_e(t)}{dt} = \frac{d\phi_1(t)}{dt} - 2\pi k_o \int_{-\infty}^{\infty} \sin(\phi_e(t)) h(t-\tau) d\tau$$

where $k_o = k_v k_m A_c A_v \rightarrow$ loop gain parameter

This is the non linear model relation equation.



Rashmi M Hullamani | Assistant Professor, Department of Telecommunication, JNNCE Shivamogga

fig: Non linear model of PLL

35

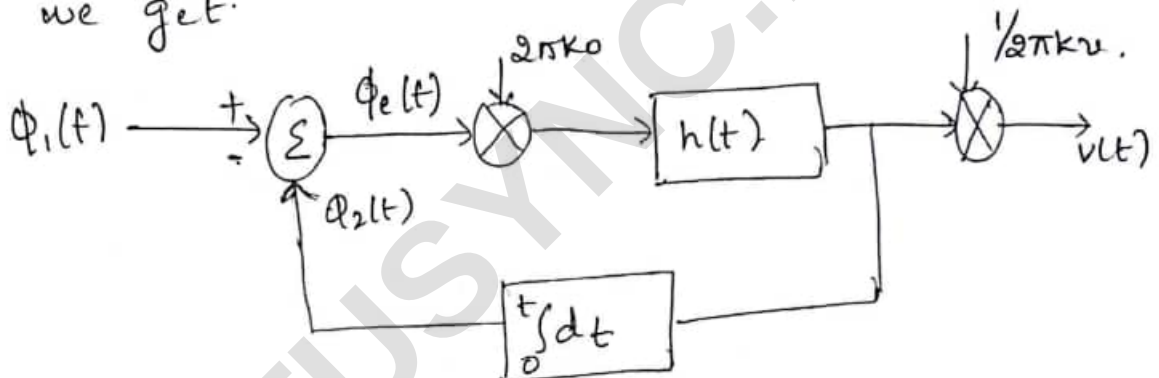
PRINCIPLES OF COMMUNICATION SYSTEMS

Linear model of PLL.

- The PLL is said to be phase lock when the phase error $\phi_e(t)$ is zero.

~~The~~ when phase error $\phi_e(t)$ is less than one radian, then $\sin[\phi_e(t)] \approx \phi_e(t)$

This approximation allows to disregard sinusoidal nonlinearity
ie eliminating sinusoidal non linearity we get.



wk.t. phase error

$$\phi_e(t) = \phi_1(t) - \phi_2(t)$$

Substituting $\phi_2(t)$
 $\phi_2(t) = 2\pi k_v \int_0^t v(t) dt$

$$\therefore \phi_e(t) = \phi_1(t) - 2\pi k_v \int_0^t v(t) dt$$

$$\therefore 2\pi k_v \int_0^t v(t) dt = \phi_1(t) - \phi_e(t)$$

assuming small error $\phi_e(t)$.

$$2\pi k_v \int_0^t v(t) dt \approx \phi_1(t)$$

PRINCIPLES OF COMMUNICATION SYSTEMS

For FM wave i/p $\phi_i(t) = 2\pi k_f \int_0^t m(t) dt$

$$\therefore 2\pi k_v \int_0^t v(t) dt = 2\pi k_f \int_0^t m(t) dt$$

Differentiating wrt t

$$k_v [v(t)] = k_f [m(t)]$$

$$v(t) = \frac{k_f}{k_v} m(t)$$

$$\therefore v(t) \propto m(t)$$

Thus o/p $v(t)$ of low pass filter is proportional to original message signal.

PRINCIPLES OF COMMUNICATION SYSTEMS

Super Heterodyne Receiver:

In communication systems, receiver not only does demodulation, but also has to perform other functions like.

* Carrier frequency Tuning:

To select desired signals like radio/TV

* Filtering:

Separates the desired signal from modulated signals.

* Amplification:

To compensate for loss of signal power during transmission.

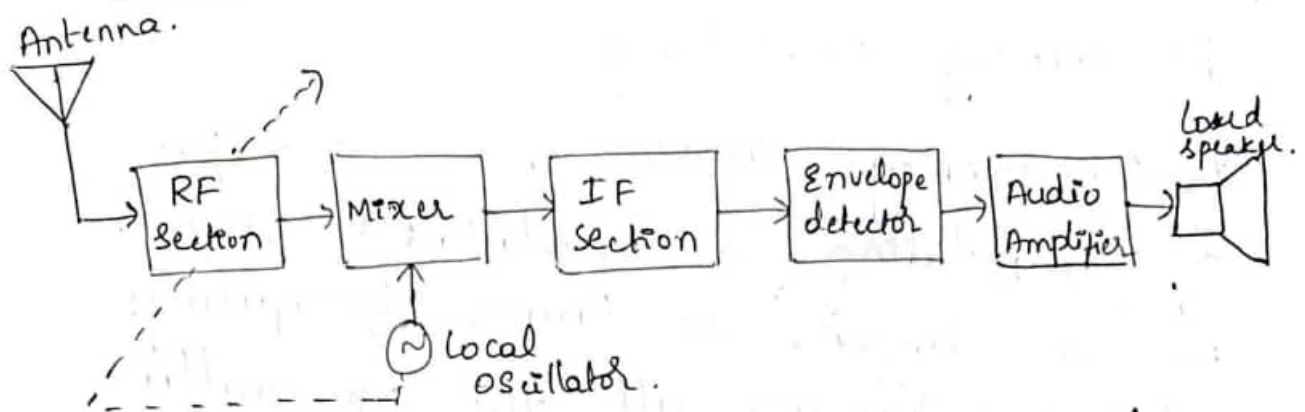
So a specified type of receiver is used in practical is superheterodyne / Superhet.

It overcomes the difficulty of having to build tunable high Q-filter.

It consists of *

- 1) Radio frequency section.
- 2) Mixer & local oscillator.
- 3) Intermediate frequency IF section.
- 4) Demodulator
- 5) power amplifier [Audio frequency]

PRINCIPLES OF COMMUNICATION SYSTEMS



Basic elements of an AM superheterodyne receiver.

- The incoming amplitude modulated wave is picked by receiving antenna and amplified in RF section that is tuned to carrier frequency of incoming wave.
- The combination of mixer & local oscillator provides heterodyning function.
i.e. incoming signal is converted to a fixed intermediate frequency i.e. $f_{IF} = f_{RF} - f_{LO}$.
where f_{LO} = frequency of local oscillator.
 f_{RF} = frequency of carrier of incoming RF wave
- "IF section" consists of one or more stages of tuned amplification. The o/p is applied to a demodulator to recover baseband signal. Finally a power amplifier is used to increase the strength of audio signal and fed to loud speaker.

Principles of Communication System

Frequency conversion:

- * Frequency conversion is the process of translating a modulated signal to a higher or lower frequency while retaining all the originally transmitted information.
- * In radio receivers, high-frequency radio signals are regularly converted to a lower, intermediate frequency, where improved gain and selectivity can be obtained. This is called down conversion.
- * In satellite communications, the original signal is generated at a low frequency and then converted to a higher frequency for transmission. This is called up conversion.

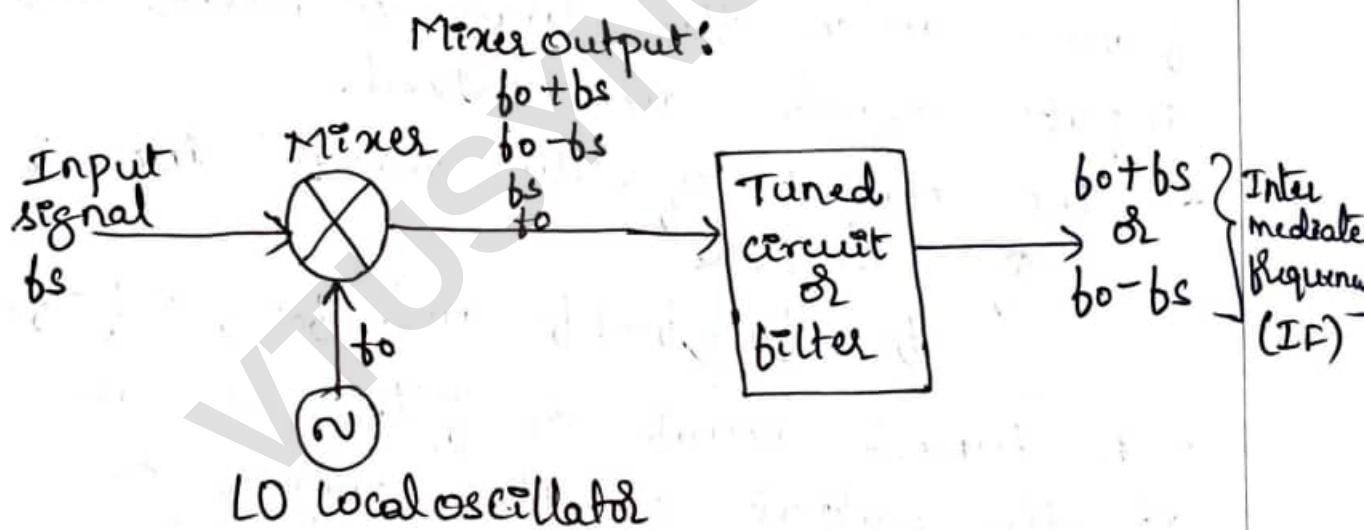
Principles of Communication System

Mixing principles:

* Frequency conversion is a form of amplitude modulation or analog multiplication carried out by a mixer circuit or converter.

* The function performed by the mixer is called heterodyning.

* The diagram of a mixer circuit is as shown below



* Mixers accept two inputs.

* The signal f_s , which is to be translated to another frequency, is applied to one input, and the sine wave from the local oscillator f_o is applied to the other input.

Principles of Communication System

* The signal to be translated can be a simple sine wave or any complex modulated signal containing sidebands.

* The oscillator is the carrier, and the signal to be translated is the modulating signal.

* The output of the mixer ^{contains} not only the carrier signal but also sidebands formed when the local oscillator and input signal are mixed.

* The output of the mixer consists of signals f_s , f_o , $f_o + f_s$ and $f_o - f_s$ or $f_s - f_o$.

* A tuned circuit or filter is used at the output of the mixer to select the desired signal and reject all the others.

Principles of Communication System

FET Mixers:

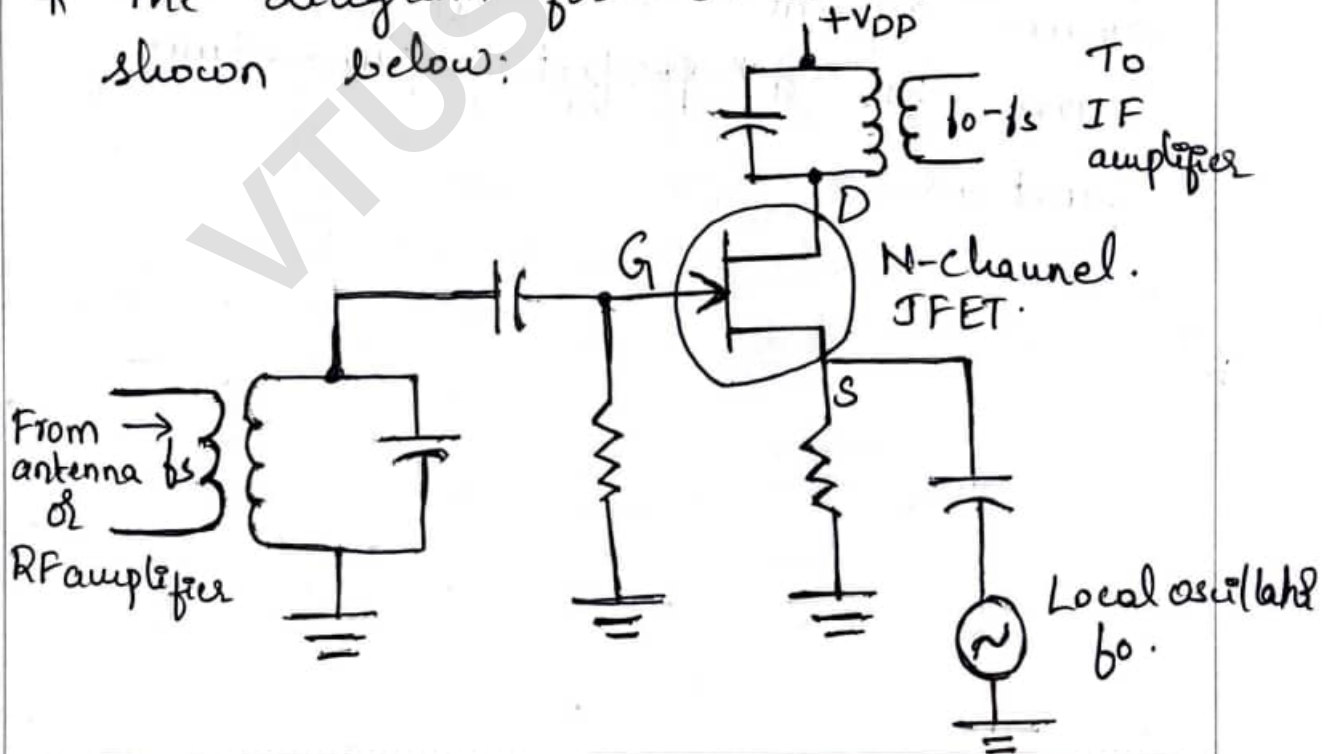
- * FETs make good mixers, because they provide
 - gain
 - low noise and
 - offer a nearly perfect square-law response.

* Two types of FET mixers

- ** ① JFET mixer
- ② MOSFET mixer.

JFET mixer:

- * The diagram for JFET mixer is as shown below:



Principles of Communication System

* The FET mixer is biased so that it operates in the nonlinear portion of its range.

* The input signal is applied to the gate, and the local oscillator signal is coupled to the source.

* Again, the tuned circuit in the drain selects the difference frequency.

* One of the best reasons for using a FET mixer is that its characteristic drain current versus gate voltage curve is a perfect square-law function.